A two-stage water treatment pilot

The pilot plant treats river water in two stages before it reaches the storage tank. The first stage removes sand and large particles, and the second stage removes fine particles and germs. Both stages were built from standard parts so that the plant can be copied by small towns. The rest of this note explains the layout of each stage and the sensors that watch it.

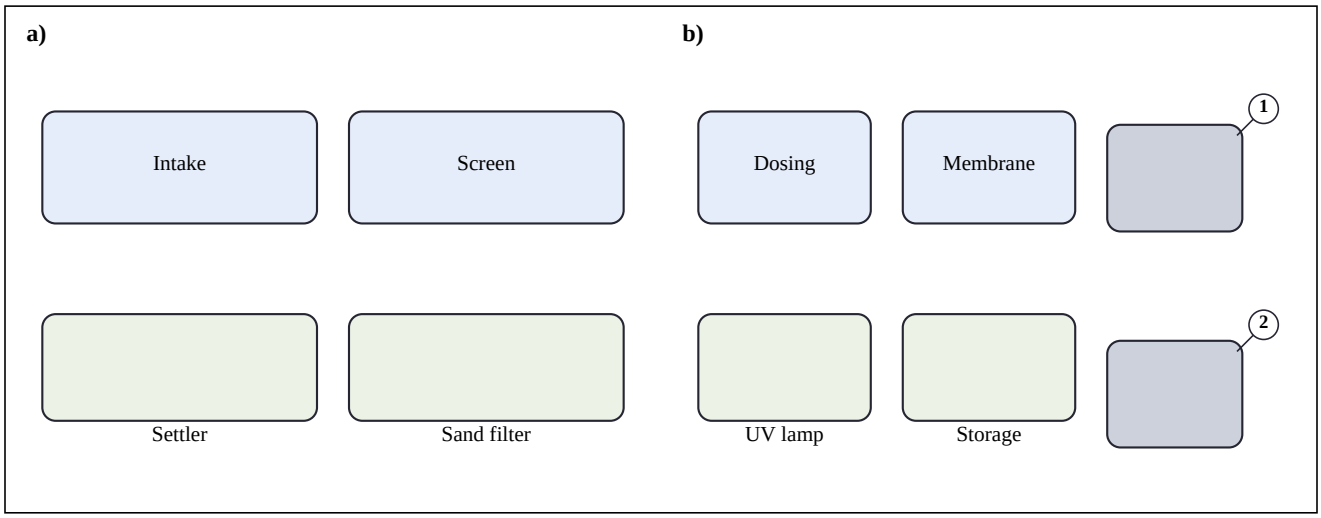

Figure 1: The two treatment stages. Stage one is shown on the left and stage two on the right.

Water is pumped from the river through a coarse intake and a rotating screen. The settler gives heavy particles about forty minutes to sink before the water moves on. A slow sand filter then removes most of the remaining turbidity. The second stage adds a small dose of coagulant, which helps the membrane catch fine particles.

An ultraviolet lamp treats the water just before it enters the storage tank. Two unnamed vessels in the second stage hold backwash water and are marked by numbered callouts. Operators drain these vessels once a day and log the volume they remove.

Sensor positions

Four sensors watch the plant, and each one is marked with a numbered callout in the sensor drawing. The turbidity probe sits after the sand filter and is the most important single reading. A pressure gauge across the membrane shows when the membrane needs cleaning. The other two sensors measure the lamp intensity and the level of the storage tank.

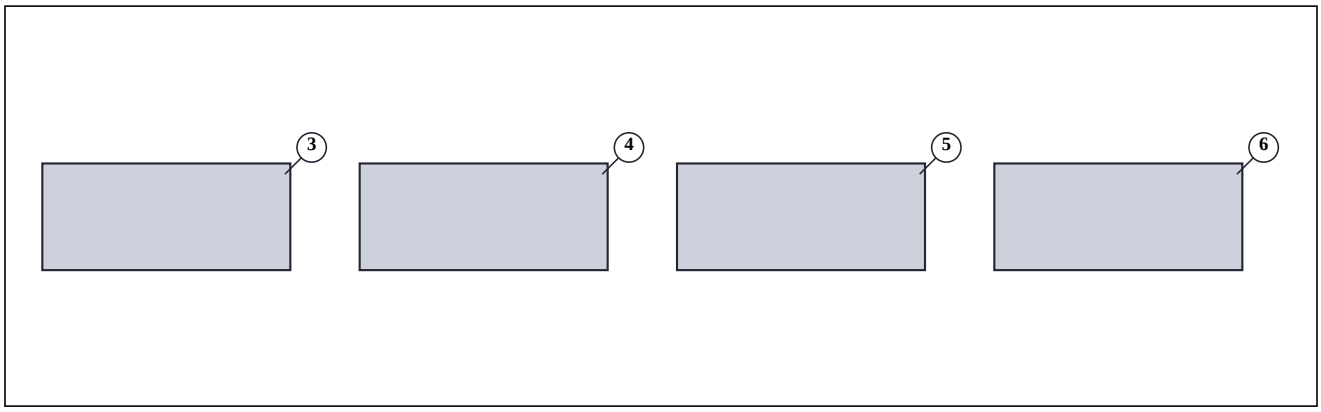

Figure 2: Sensor drawing. Numbers mark the four sensors described in the text.

Table 1: Alarm limits used during the pilot.

Sensor	Low	High	Action
Turbidity	none	1.0 NTU	stop outflow
Membrane pressure	0.2 bar	1.5 bar	clean
Lamp intensity	60 %	none	replace lamp
Tank level	10 %	95 %	adjust pump

Alarm limits were set conservatively for the first season and will be relaxed once more data is available. Every alarm sends a text message to the operator on duty and is written to the plant log. During the pilot the turbidity alarm fired six times, always after heavy rain upstream.